

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

Kopiereg voorbehou

Examination

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Module EBN111
Elektrisiteit en Elektronika
21 Junie 2016

Module EBN111
Electricity and Electronics
21 June 2016



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: Exam information:

Maksimum punte: Maximum marks:	90	Volpunte: Full marks:	93
Duur van vraestel: Duration of paper:	180 minute 180 minutes	Oopboek / toeboek: Open / closed book:	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (including this page):		28 (Antwoordblad uigesluit / Answer sheet excluded)	

BELANGRIK- IMPORTANT

- The examination regulations of the University of Pretoria apply.*
Die eksamenregulasies van die Universiteit van Pretoria geld.
- All questions must be answered on the ANSWER SHEET.*
Alle vrae moet op die ANTWOORDBLAD beantwoord word.
- The English text of this question paper will be given precedence in the interpretation of the content.*
Die Engelse teks van hierdie vraestel sal voorkeur geniet met betrekking tot die interpretasie van die inhoud.
- Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
- Answers must include units!*
Antwoorde moet eenhede insluit!
- Only non-programmable calculators may be used.*
Slegs nie-programmeerbare sakrekenaars mag gebruik word.
- Polar form:** $A \angle \theta$; **Rectangular form:** $a + jb$
Polêre vorm: $A \angle \theta$; **Reghoekige vorm:** $a + jb$
- Final answers must be written as real numbers, and not as fractions.**
Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Interne Eksaminator(e): Internal examiner(s):	D. le Roux, D. Bhatt & G. Hancke
Eksterne eksaminator: External examiner:	J. Joubert

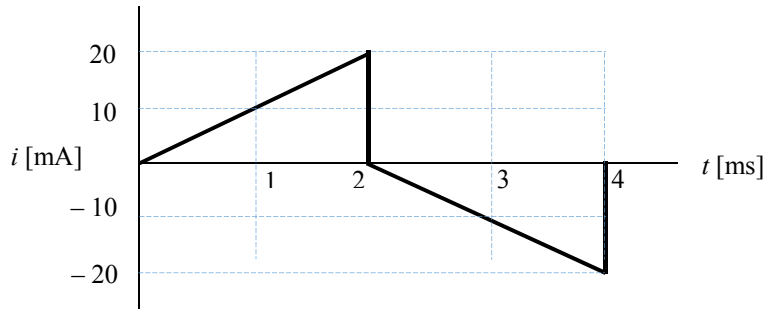
QUESTION/VRAAG 1 [33]

Question 1.1 [4]

- a) A kettle is powered by 250 V and draws a current of 10 A for 3 minutes to boil an amount water. If energy is sold at R2,00/kWh, how much does it cost to boil this amount of water (in cent)?
- b) The current flowing through a $100\ \Omega$ resistor is shown in the graph below. Calculate the charge that flows through the resistor between $t = 1\text{ ms}$ and $t = 4\text{ ms}$.

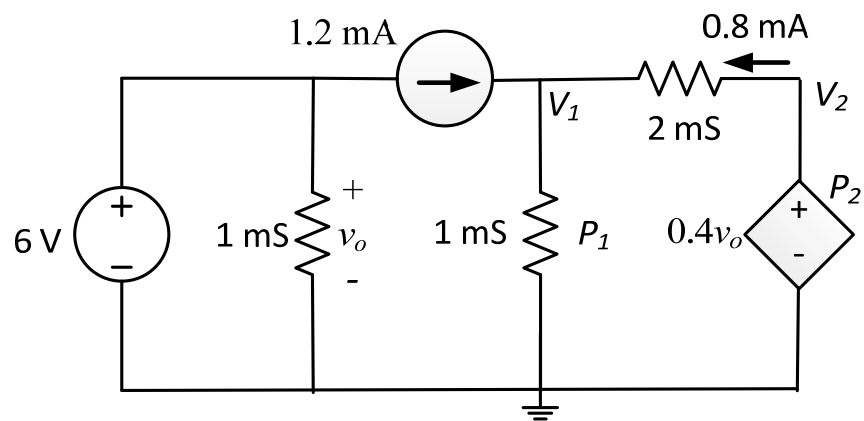
Vraag 1.1 [4]

- a) 'n Ketel is gekoppel aan 'n bron van 250 V en benodig 10 A vir 3 minute om 'n hoeveelheid water te kook. Indien energie verkoop word teen R2,00/kWh, hoeveel kos dit om hierdie hoeveelheid water te kook (in sent)?
- b) Die stroom wat deur 'n $100\ \Omega$ weerstand vloei word in die onderstaande grafiek aangedui. Bereken die lading wat deur die weerstand vloei tussen $t = 1\text{ ms}$ en $t = 4\text{ ms}$.



Question 1.2 [4]
Determine P_1 and P_2 .

Vraag 1.2 [4]
Bepaal P_1 en P_2 .

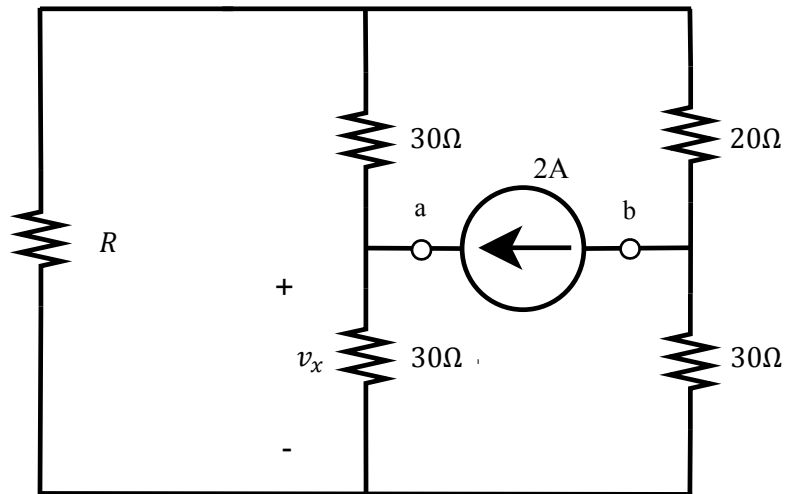


Question 1.3 [6]

- (a) If $R = \infty \Omega$ calculate v_x .
(b) If $R = 0 \Omega$ calculate v_x .
(c) If $R = 0 \Omega$ calculate the total resistance as seen from the terminals of the current source.

Vraag 1.3 [6]

- (a) Indien $R = \infty \Omega$ bereken v_x .
(b) Indien $R = 0 \Omega$ bereken v_x .
(c) Indien $R = 0 \Omega$ bereken die totale weerstand soos gesien vanaf die terminale van die stroombron.



Question 1.4 [4]

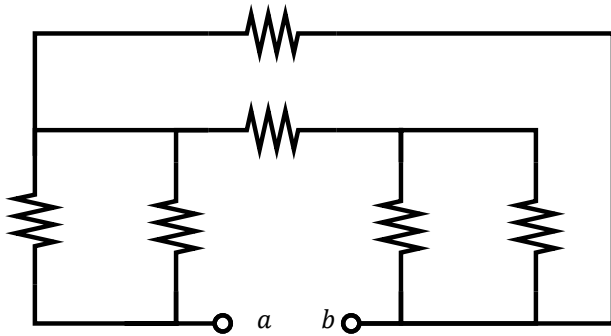
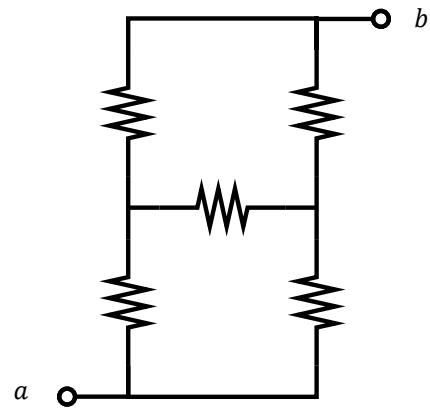
Determine the equivalent resistance of the two circuits below with respect to terminals a and b . Assume:

- All resistors in circuit 1 have a value of $20\ \Omega$.
- All resistors in circuit 2 have a value of $75\ \Omega$.

Vraag 1.4 [4]

Bepaal die ekwivalente weerstand van onderstaande twee kringbane ten opsigte van terminale a en b . Aanvaar:

- Alle weerstande in kringbaan 1 het 'n waarde van $20\ \Omega$.
- Alle weerstande in kringbaan 2 het 'n waarde van $75\ \Omega$.

Kringbaan / Circuit 1**Kringbaan / Circuit 2**

Question 1.5 [4]

Use mesh analysis to formulate two equations of the following form:

$$ai_1 - 5i_2 = b$$

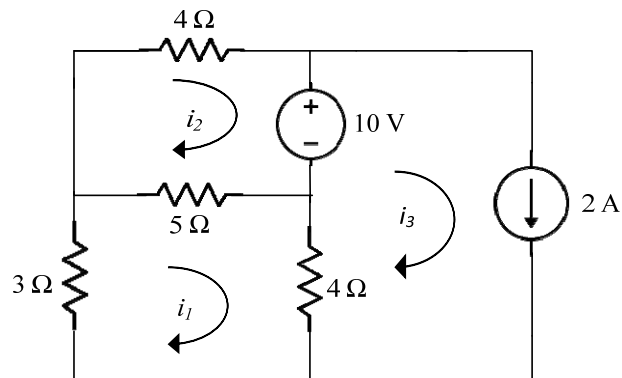
$$ci_1 + di_2 = 10$$

Vraag 1.5 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$ai_1 - 5i_2 = b$$

$$ci_1 + di_2 = 10$$



Question 1.6 [4]

Use nodal analysis to set up two equations of the following form:

$$eV_2 + 3V_3 = f$$

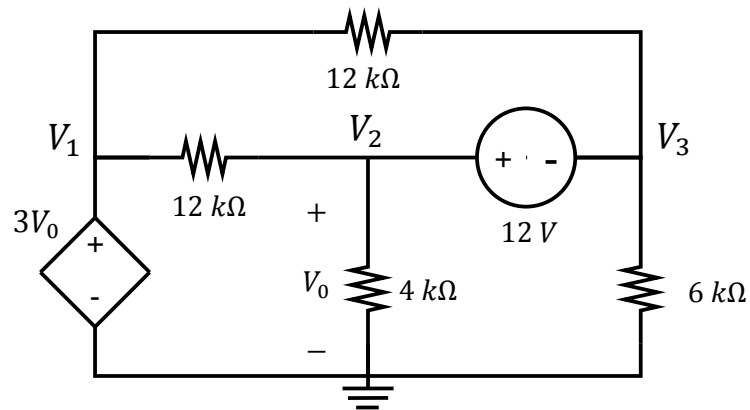
$$gV_2 + hV_3 = 12$$

Vraag 1.6 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$eV_2 + 3V_3 = f$$

$$gV_2 + hV_3 = 12$$



Question 1.7 [4]

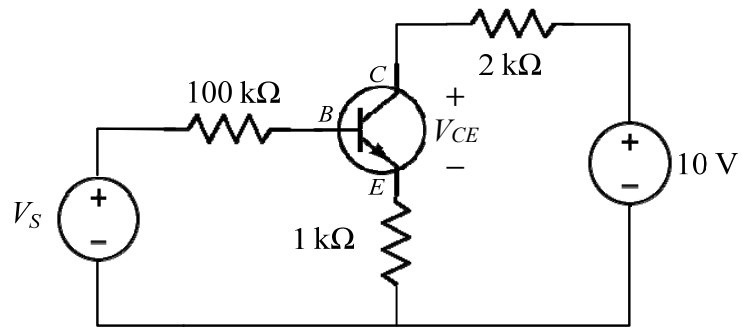
Given: $\beta = 99$ and $V_{BE} = 0.7$ V.

- a) If $I_E = 3$ mA, determine V_S . [2]
b) If $I_B = 15$ μ A, determine V_{CE} . [2]

Vraag 1.7 [4]

Gegee: $\beta = 99$ en $V_{BE} = 0.7$ V.

- a) Indien $I_E = 3$ mA, bepaal V_S . [2]
b) Indien $I_B = 15$ μ A, bepaal V_{CE} . [2]

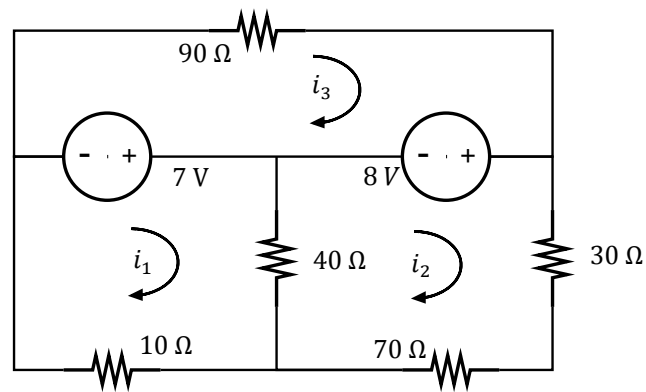


Question 1.8 [3]

Use inspection to set up a matrix equation in terms of the mesh currents.

Vraag 1.8 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die lusstrome.



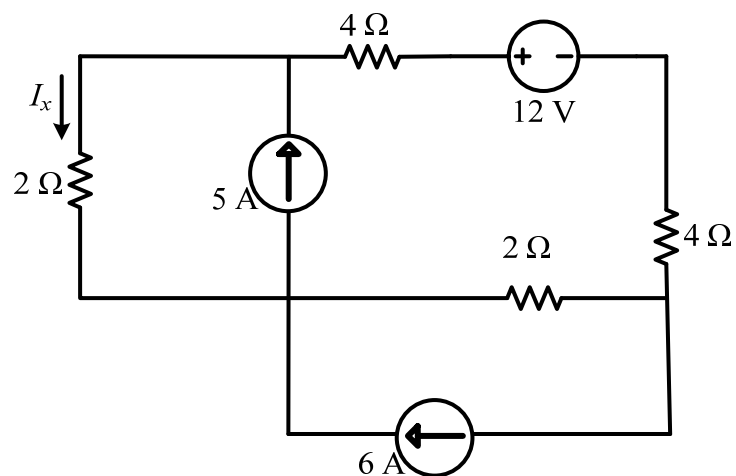
VRAAG/QUESTION 2 [28]

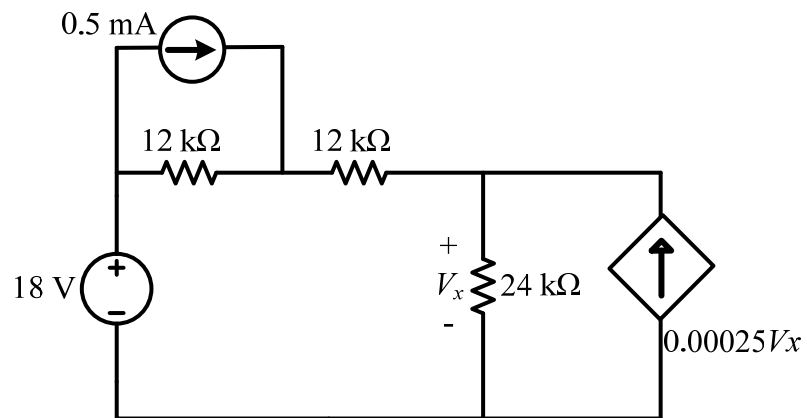
Question 2.1 [4]

- a) Determine I_x due to 12 V voltage source alone.
- b) Determine I_x due to 6 A current source alone.

Vraag 2.1 [4]

- a) Bepaal I_x slegs as gevolg van die 12 V spanningsbron.
- b) Bepaal I_x slegs as gevolg van die 6 A stroombron.



Question 2.2 [2]Use source transformation to determine the voltage, V_x .**Vraag 2.2 [2]**Gebruik brontransformatie om die spanning V_x te bepaal.

Question 2.3 [4]

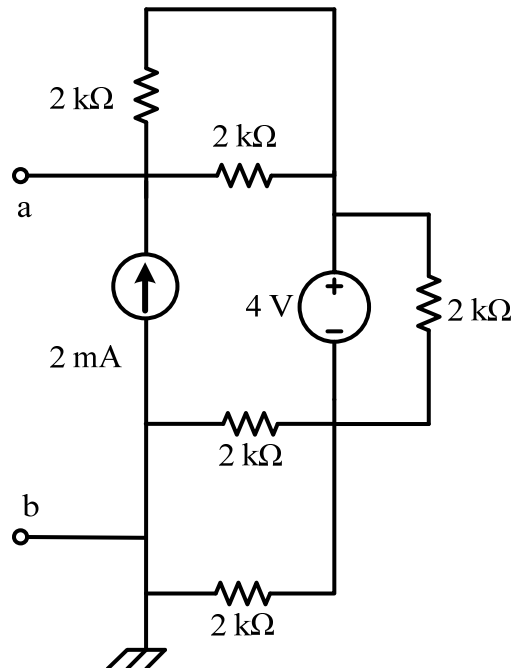
Calculate the Thevenin equivalent with respect to terminals a - b :

- a) V_{TH} and
- b) R_{TH}

Vraag 2.3 [4]

Bereken die Thevenin ekwivalent met verwysing tot terminale a en b .

- a) V_{TH} en
- b) R_{TH}

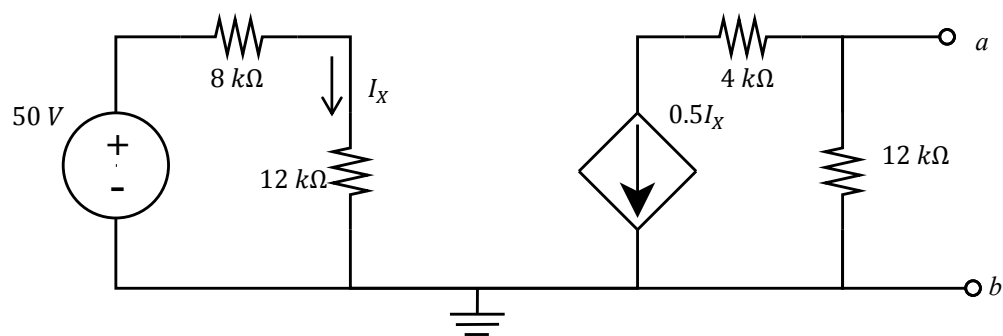


Question 2.4 [4]

Determine the Norton equivalent with respect to terminals a and b .

Vraag 2.4 [4]

Bepaal die Norton ekwivalent met betrekking tot terminale a en b .

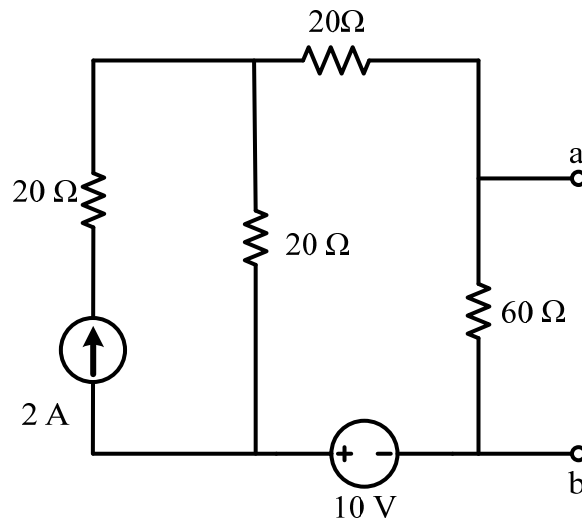


Question 2.5 [4]

- a) What is the maximum power, P_{max} , which can be delivered to a load if the load is connected to a circuit with a Norton equivalent current of 5.5 A and Norton equivalent resistance of 60Ω ?
- b) For the circuit below, what is the power P which can be delivered to a 50Ω load if the load is connected between terminals a and b ?
(The Thevenin equivalent voltage of the circuit below is 30 V. The 50Ω load is not shown in the circuit.)

Vraag 2.5 [4]

- a) Wat is die maksimum drywing, P_{max} , wat aan 'n las voorsien kan word indien die las aan 'n kringbaan gekoppel word met 'n Norton ekwivalente stroom van 5.5 A en 'n Norton ekwivalente weerstand van 60Ω ?
- b) Vir onderstaande kringbaan, wat is die drywing P wat aan 'n 50Ω las voorsien kan word indien die las tussen terminale a en b gekoppel word?
(Die Thevenin ekwivalente spanning is 30 V. Die 50Ω word nie in die kringbaan aangedui nie.)

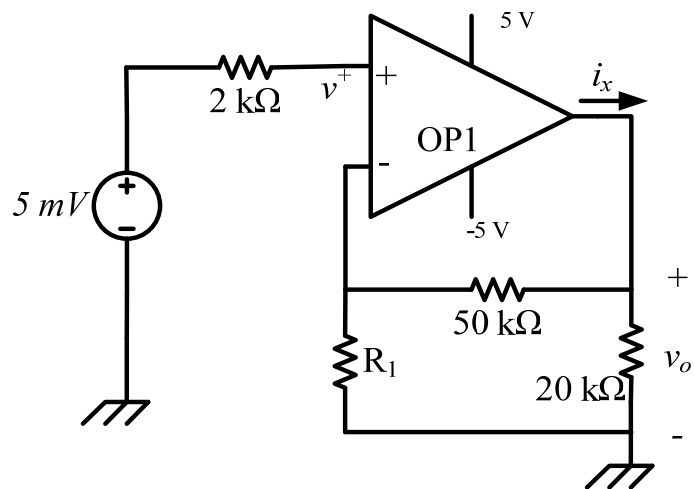


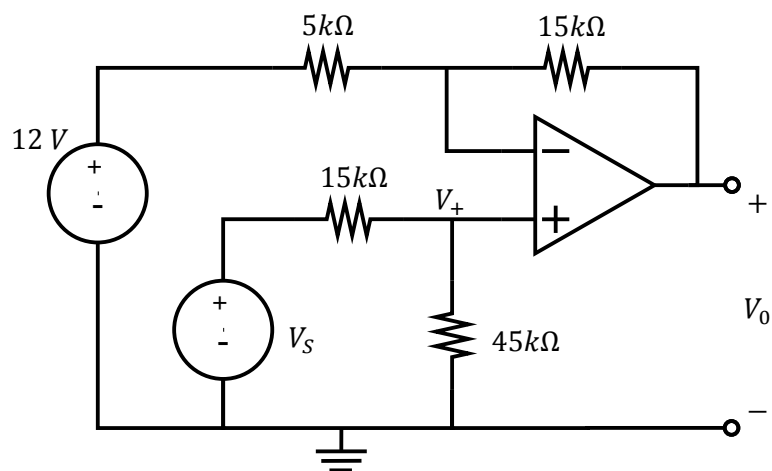
Question 2.6 [4]

- a) Calculate the current, i_x , when $R_I = 5 \text{ k}\Omega$.
b) Given $|V_{CC}| = 5 \text{ V}$. For what value of R_I will the output voltage v_o of the operational amplifier circuit reach its maximum saturation point?

Vraag 2.6 [4]

- a) Bereken die stroom i_x indien $R_I = 5 \text{ k}\Omega$.
b) Gegee $|V_{CC}| = 5 \text{ V}$. Vir watter waarde van R_I sal die uitsetspanning v_o van die operasionele versterkerkringbaan sy maksimum versadigingspunt bereik?



Question 2.7 [4]Determine V_0 and V_S if $V_+ = 6V$.**Vraag 2.7 [4]**Bepaal V_0 en V_S indien $V_+ = 6V$.

Question 2.8 [4]

Calculate

- a) v_1 ,
- b) v_2 ,
- c) v_3 and
- d) v_o

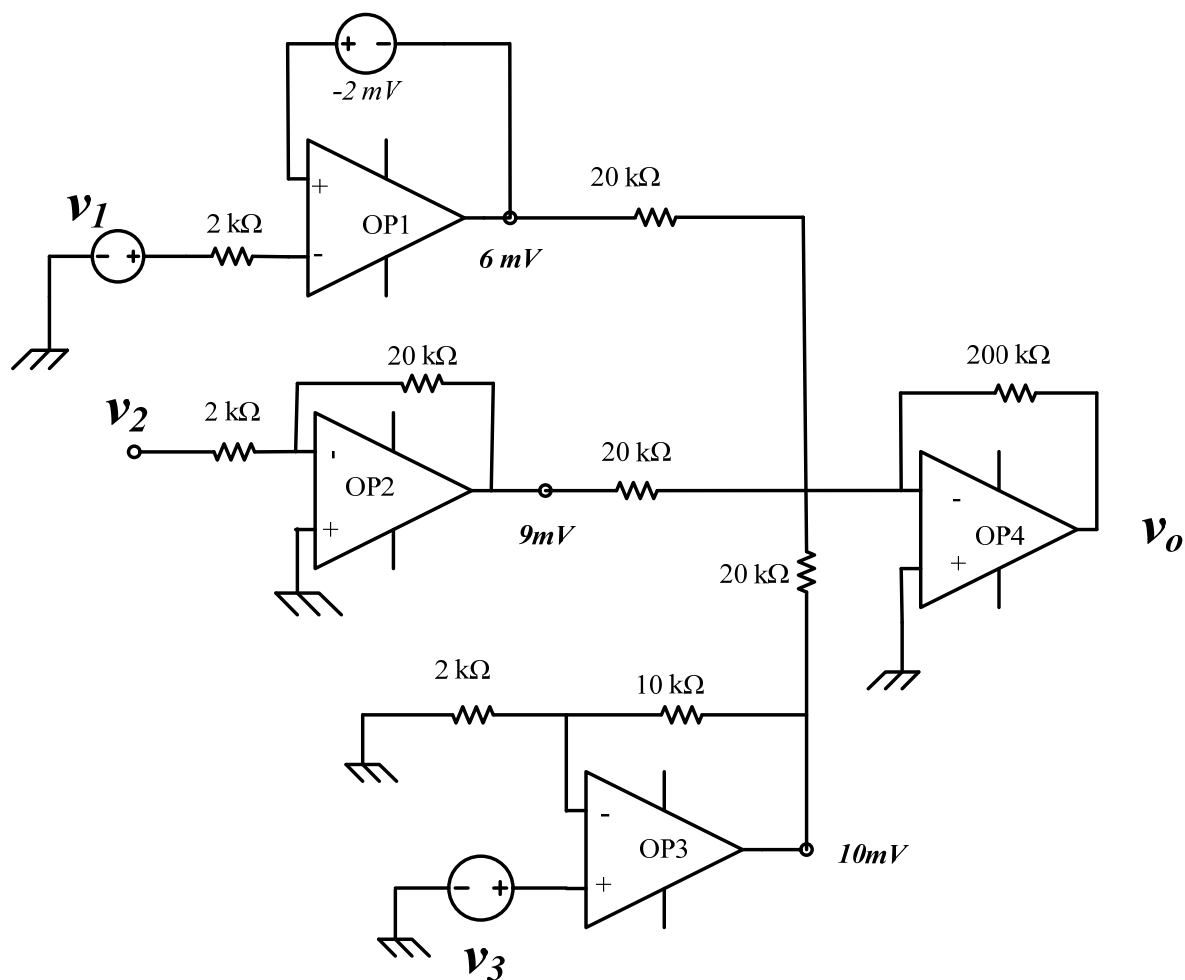
(The answers are independent of each other.)

Vraag 2.8 [4]

Bereken

- a) v_1 ,
- b) v_2 ,
- c) v_3 en
- d) v_o

(Die antwoorde is onafhanklik van mekaar.)



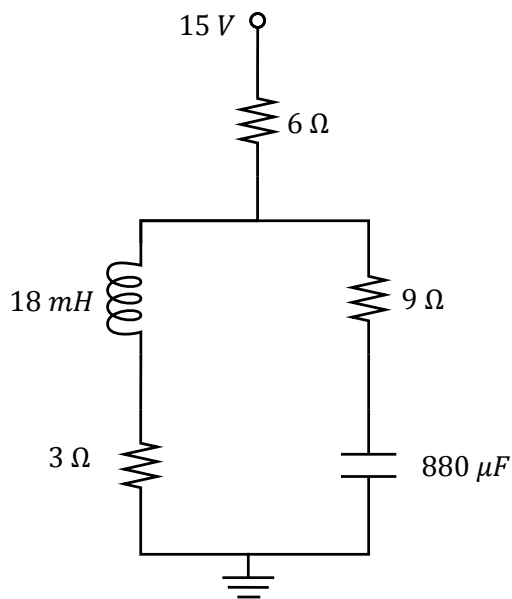
VRAAG/QUESTION 3 [30]

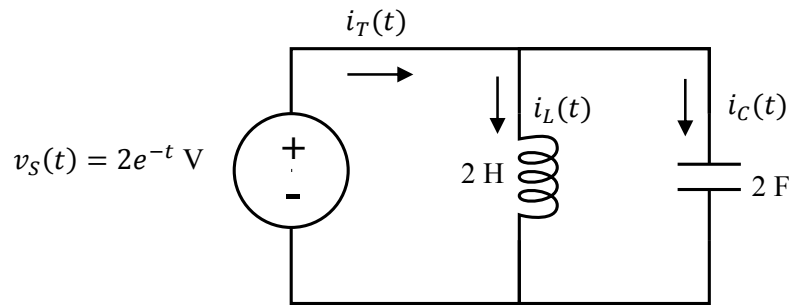
Question 3.1 [4]

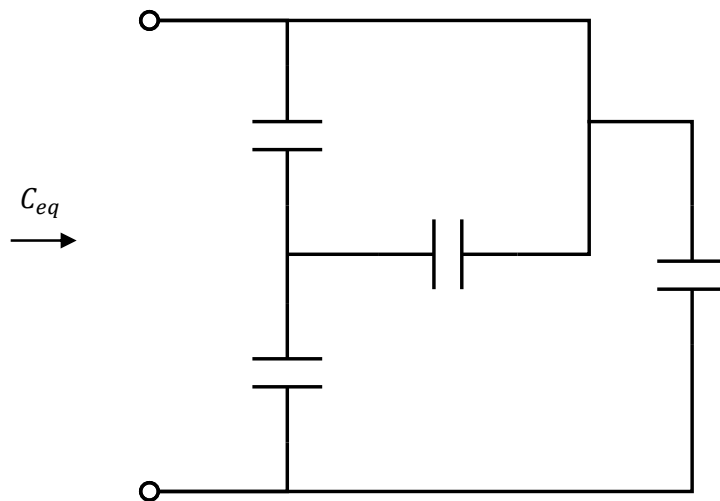
Assume steady-state (direct current) conditions and determine the energy stored in the capacitor and inductor respectively.

Vraag 3.1 [4]

Aanvaar gestadigde toestand (gelykstroom) en bereken die energie gestoor in die kapasitor en induktor respektiewelik.



Question 3.2 [2]Calculate $i_T(t)$ if $v_S(t) = 2e^{-t}$ V.Assume $i_C(0) = 0$ A and $i_L(0) = 0$ A.**Vraag 3.2 [2]**Bereken $i_T(t)$ indien $v_S(t) = 2e^{-t}$ V.Aanvaar $i_C(0) = 0$ A and $i_L(0) = 0$ A.

Question 3.3 [2]Calculate C_{eq} if all capacitors are $15\ \mu\text{F}$.**Vraag 3.3 [2]**Bereken C_{eq} indien alle kapasitors $15\ \mu\text{F}$ is.

Question 3.4 [4]

- a) Which of the following signals is leading and by how many degrees:

$$\bar{V}_1 = 8\angle 75^\circ$$

$$i_2(t) = 10 \sin(\omega t - 110^\circ)$$

- b) Simplify $v(t)$ and write as **a single sinusoid**

$$v(t) = V_m \cos(\omega t + \theta^\circ) \text{ with } V_m > 0 \text{ and } -180^\circ \leq \theta \leq 180^\circ.$$

$$v(t) = 7 \cos(\omega t + 51.7868^\circ) + 5 \sin(\omega t - 60^\circ) \text{ V}$$

Vraag 3.4 [4]

- a) Watter van die volgende seine loop voor en met hoeveel grade:

$$\bar{V}_1 = 8\angle 75^\circ$$

$$i_2(t) = 10 \sin(\omega t - 110^\circ)$$

- b) Vereenvoudig $v(t)$ en skryf as **enkele sinusied**

$$v(t) = V_m \cos(\omega t + \theta^\circ) \text{ met } V_m > 0 \text{ en } -180^\circ \leq \theta \leq 180^\circ.$$

$$v(t) = 7 \cos(\omega t + 51.7868^\circ) + 5 \sin(\omega t - 60^\circ) \text{ V}$$

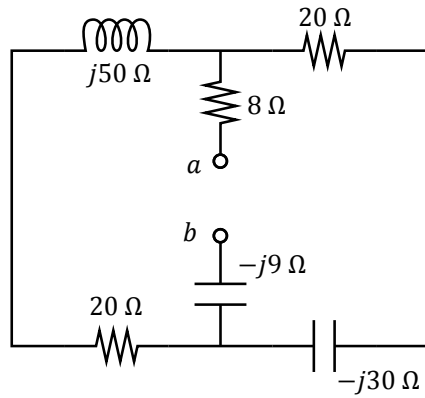
Question 3.5 [2] A $120\ \Omega$ resistor and a 40 mH inductor are connected in series. This series combination is driven by a sinusoidal voltage source whose voltage is $25 \cos(4000t + 23.13^\circ)\text{ V}$. Find an expression for the current $i(t)$ flowing through the circuit. (Assume that you may use the positive form of Ohm's law).	Vraag 3.5 [2] 'n $120\ \Omega$ resistor en 'n 40 mH induktor word in serie gekonnekteer. Hierdie serie kombinasie word deur 'n sinusvormige spanningsbron van $25 \cos(4000t + 23.13^\circ)\text{ V}$ aangedryf. Vind 'n uitdrukking vir die stroom $i(t)$ wat deur die kringbaan vloei. (Gebruik die positiewe vorm van Ohm se Wet.)
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Question 3.6 [2]

Determine the total impedance Z_{ab} between terminals a and b . Write the answer in **rectangular form**.

Vraag 3.6 [2]

Bereken die totale impedansie Z_{ab} tussen terminale a en b . Skryf die antwoord in **reghoekige formaat**.

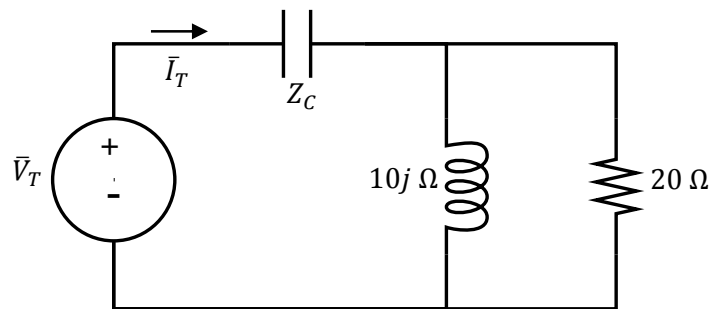


Question 3.7 [2]

In the figure below the capacitor is adjusted until the voltage phasor $\bar{V}_T$ is in phase with the current phasor $\bar{I}_T$. Specify the required impedance of the capacitor where this happens. (I.e. determine Z_C such that the phase of $\bar{V}_T$ and $\bar{I}_T$ differ by 0° .)

Vraag 3.7 [2]

In onderstaande figuur word die kapasitor aangepas totdat die spanningsfasor $\bar{V}_T$ en stroomfasor $\bar{I}_T$ in fase is met mekaar. Bereken die impedansie van die kapasitor waar dit gebeur. (M.a.w. bepaal Z_C sodat die fase van $\bar{V}_T$ en $\bar{I}_T$ met 0° van mekaar verskil.)



Question 3.8 [4]

Use mesh analysis to set up two equations of the following form:

$$(a_1 + jb_1)\bar{I}_1 + (a_2 + jb_2)\bar{I}_2 = j6$$

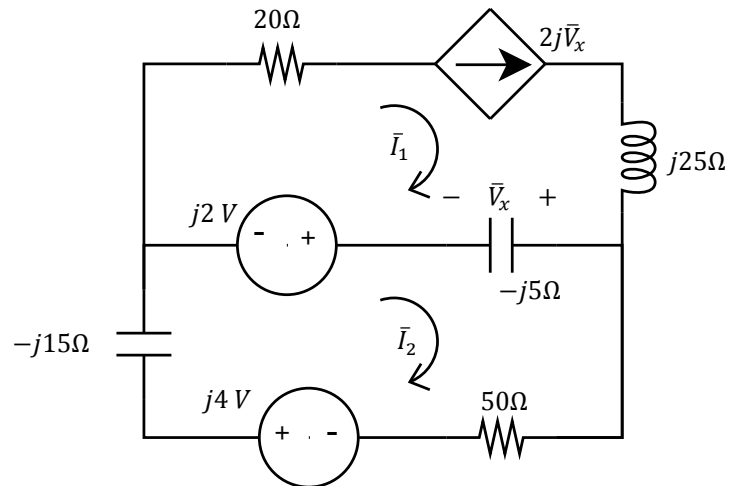
$$a_3\bar{I}_1 + a_4\bar{I}_2 = 0$$

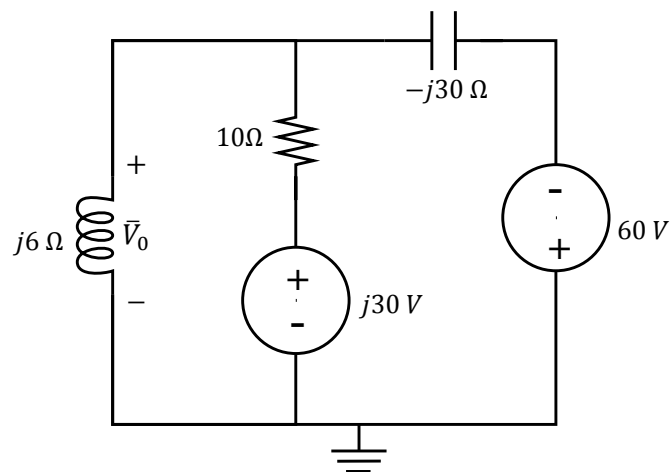
Vraag 3.8 [4]

Gebruik lusanalyse en stel twee vergelykings van die volgende vorm op.

$$(a_1 + jb_1)\bar{I}_1 + (a_2 + jb_2)\bar{I}_2 = j6$$

$$a_3\bar{I}_1 + a_4\bar{I}_2 = 0$$



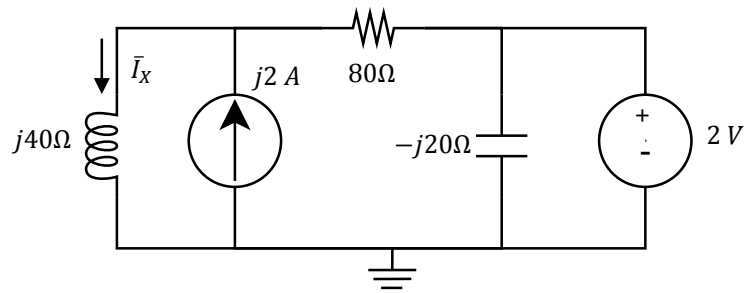
Question 3.9 [2]Determine $\bar{V}_0$. Write the answer in **rectangular form**.**Vraag 3.9 [2]**Bepaal $\bar{V}_0$. Skryf die antwoord in **reghoekige vorm**.

Question 3.10 [2]

Use superposition and determine $\bar{I}_x$ as contributed by the current source only. Write the answer in **rectangular form**.

Vraag 3.10 [2]

Gebruik superposisie om die bydra van die stroombron alleen tot die stroom $\bar{I}_x$ te bereken. Skryf die antwoord in **reghoekige vorm**.



Question 3.11 [4]

Determine the Thevenin equivalent voltage and impedance of the circuit w.r.t. terminals a and b .

a) V_{TH} in **polar** format.

b) Z_{TH} in **rectangular** format.

Question 3.11 [4]

Bepaal die Thevenin ekwivalente stroom en impedansie van die stroombaan met verwysing tot terminale a en b .

a) V_{TH} in **polêre** vorm.

b) Z_{TH} in **reghoekige** vorm.

